



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE
Batch : 2025-29

Assignment : 18

Subject: Programming with Java

ASSIGNMENT No : 18

TITLE : Write a Java GUI program to demonstrate button click event handling.

Problem Statement

Write a Java GUI program to demonstrate button click event handling.

Learning Objectives

To implement button click events using Java event handling mechanisms.

Theory

Event handling allows Java programs to respond to user actions such as button clicks, mouse movements, and keyboard input. Swing uses ActionListener to process button click events. Event-driven programming separates user interface design from event-processing logic, making applications interactive and user-friendly



Code and Output Screenshots

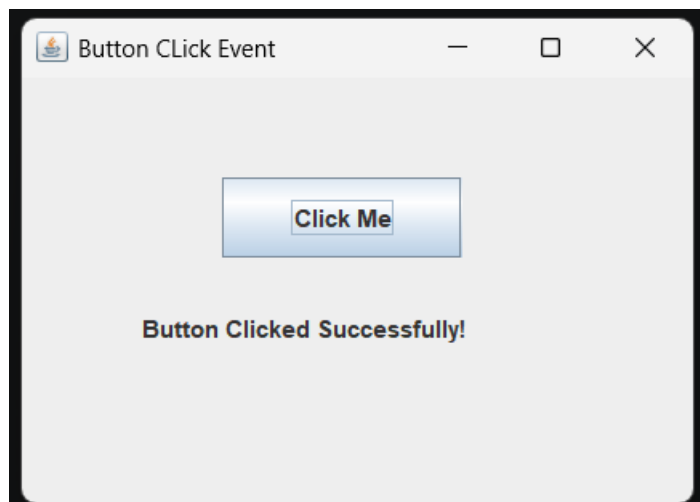
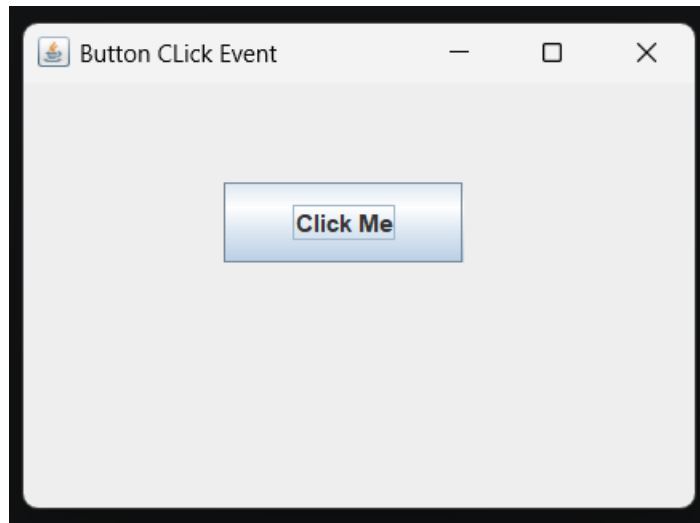
```
J A18.java > Language Support for Java(TM) by Red Hat > A18
1  import java.awt.event.ActionEvent;
2  import java.awt.event.ActionListener;
3  import javax.swing.*;
4
5  public class A18 {
6      Run main | Debug main | Run | Debug
7      public static void main (String[] args) {
8          JFrame frame = new JFrame("Button Click Event");
9          frame.setSize(350, 250);
10         frame.setLayout(null);
11         frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
12
13         JButton button = new JButton("Click Me");
14         button.setBounds(100, 50, 120, 40);
15
16         JLabel label = new JLabel();
17         label.setBounds(60, 110, 250, 30);
18
19         button.addActionListener(new ActionListener() {
20             public void actionPerformed(ActionEvent e) {
21                 label.setText("Button Clicked Successfully!");
22             }
23         });
24
25         frame.add(button);
26         frame.add(label);
27         frame.setVisible(true);
28     }
```



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Conclusion : Thus, the experiment successfully demonstrates button click event handling in Java Swing using ActionListener. The program responds to the button click and displays a message, providing a basic understanding of GUI event handling in Java.



Exercise

Q1) Create a GUI calculator where buttons perform addition and subtraction.

Sol) Code :

```
J A18E1.java > Java > A18E1
1  import javax.swing.*;
2  public class A18E1 {
    Run main | Debug main | Run | Debug
3  public static void main(String[] args) {
4
5      JFrame frame = new JFrame("Calculator");
6
7      JLabel num1Label = new JLabel("Number 1:");
8      JLabel num2Label = new JLabel("Number 2:");
9
10     JTextField num1 = new JTextField();
11     JTextField num2 = new JTextField();
12
13     JButton add = new JButton("Addition");
14     JButton subtract = new JButton("Subtraction");
15
16     num1Label.setBounds(20, 40, 80, 30);
17     num1.setBounds(100, 40, 150, 30);
18
19     num2Label.setBounds(20, 80, 80, 30);
20     num2.setBounds(100, 80, 150, 30);
21
22     add.setBounds(30, 130, 110, 30);
23     subtract.setBounds(150, 130, 120, 30);
24
25     frame.add(num1Label);
26     frame.add(num1);
27     frame.add(num2Label);
28     frame.add(num2);
29     frame.add(add);
30     frame.add(subtract);
31
32     frame.setSize(320, 220);
33     frame.setLayout(null);
34     frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
35     frame.setVisible(true);
36
37     add.addActionListener(e -> {
38         int a = Integer.parseInt(num1.getText());
39         int b = Integer.parseInt(num2.getText());
40         JOptionPane.showMessageDialog(frame, "Result: " + (a + b));
41     });
42
43     subtract.addActionListener(e -> {
44         int a = Integer.parseInt(num1.getText());
45         int b = Integer.parseInt(num2.getText());
46         JOptionPane.showMessageDialog(frame, "Result: " + (a - b));
47     });
48 }
```



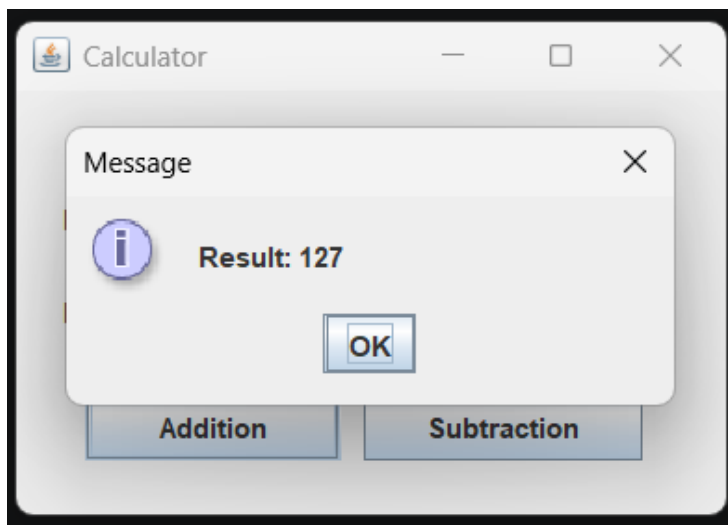
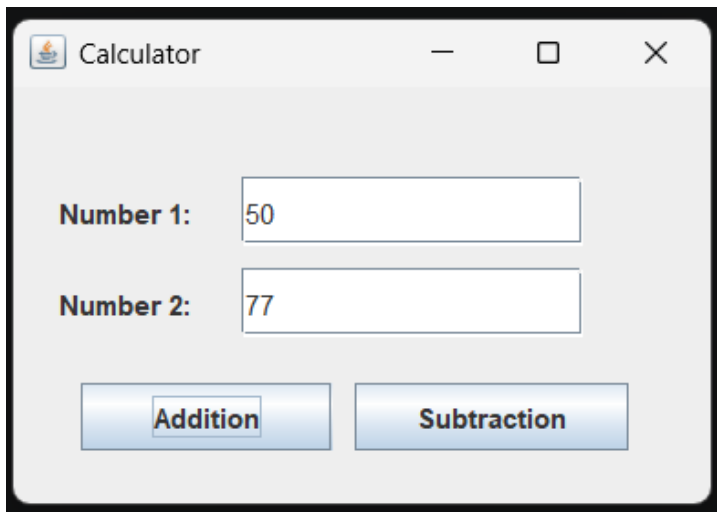
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Output :

Case 1 : Addition



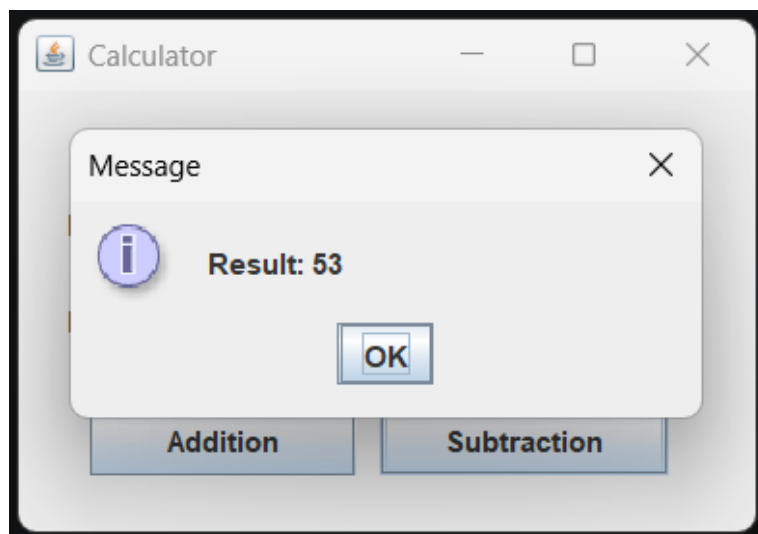
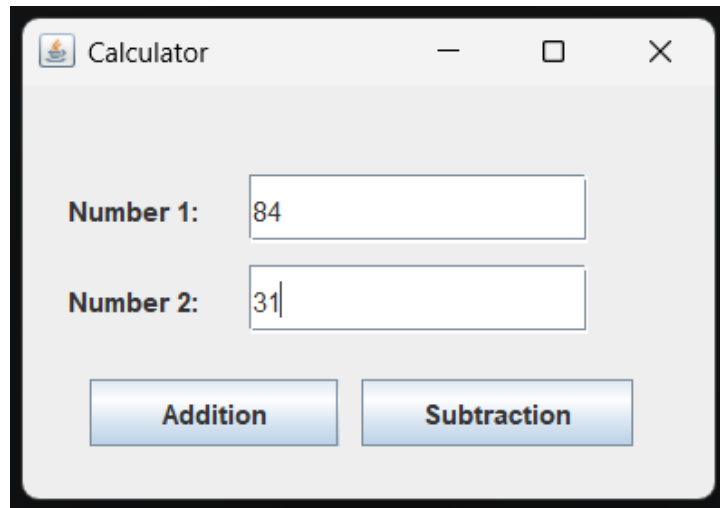


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Case 2 : Subtraction





Q2) Create a Bank Balance Calculator GUI application using Java Swing where the user enters the initial balance and transaction amount. Use buttons to perform addition (deposit) and subtraction (withdrawal) and display the updated balance.

Sol) Code :

```
J A18E2.java > Java > A18E2
1  import javax.swing.*;
2  public class A18E2 {
    Run main | Debug main | Run | Debug
3      public static void main(String[] args) {
4
5          JFrame f = new JFrame("Bank Balance Calculator");
6
7          JLabel bLabel = new JLabel("Initial Balance:");
8          JLabel aLabel = new JLabel("Amount:");
9
10         JTextField bField = new JTextField();
11         JTextField aField = new JTextField();
12
13         JButton deposit = new JButton("Deposit");
14         JButton withdraw = new JButton("Withdraw");
15
16         bLabel.setBounds(30, 40, 120, 30);
17         bField.setBounds(150, 40, 180, 30);
18
19         aLabel.setBounds(30, 90, 120, 30);
20         aField.setBounds(150, 90, 180, 30);
21
22         deposit.setBounds(70, 150, 100, 30);
23         withdraw.setBounds(200, 150, 100, 30);
24
25         f.add(bLabel);
26         f.add(bField);
27         f.add(aLabel);
28         f.add(aField);
29         f.add(deposit);
30         f.add(withdraw);
31
32         f.setSize(380, 250);
33         f.setLayout(null);
34         f.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
35         f.setVisible(true);
36
37         deposit.addActionListener(e -> {
38             double balance = Double.parseDouble(bField.getText());
39             double amount = Double.parseDouble(aField.getText());
40
41             JOptionPane.showMessageDialog(f,
42                 "Updated Balance: " + (balance + amount));
43         });
44
45         withdraw.addActionListener(e -> {
46             double balance = Double.parseDouble(bField.getText());
47             double amount = Double.parseDouble(aField.getText());
48
49             JOptionPane.showMessageDialog(f,
50                 "Updated Balance: " + (balance - amount));
51         });
52     }
```



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Output :

Case 1 : Deposit

The screenshot shows a window titled "Bank Balance Calculator". It contains two input fields: "Initial Balance:" with the value "85000" and "Amount:" with the value "3200". Below these fields are two buttons: "Deposit" and "Withdraw".

The screenshot shows the same "Bank Balance Calculator" window, but with a message dialog box overlaid. The dialog box is titled "Message" and contains an information icon (i) and the text "Updated Balance: 88200.0". There is an "OK" button at the bottom of the dialog box. The "Deposit" and "Withdraw" buttons are still visible in the background.



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Case 2 : Withdraw

Bank Balance Calculator

Initial Balance: 6700

Amount: 391

Deposit Withdraw

Bank Balance Calculator

Message

Updated Balance: 6309.0

OK

Deposit Withdraw